

Submitted on Behalf of Modine Manufacturing

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At Modine, we are Engineering a Cleaner, Healthier World™. Building on more than 100 years of excellence in thermal management, we provide trusted systems and solutions that improve air quality and conserve energy and natural resources. Our employees are at work in every corner of the globe, delivering the solutions our customers need, where they need them. While we believe that the United States is well positioned to be the preeminent country to achieve Artificial Intelligence (“AI”) dominance, we believe there are certain challenges that must be addressed to ensure that result. Our long tenured experience providing solutions to data center customers of different shapes and size all over the world has informed our opinion in this RFI.

We believe there are four critical pillars required to support development, implementation and control of the race for AI dominance. Specifically, we believe the following four pillars are essential:

1. Best in class chip and board technology
2. Power availability and reliability
3. Comprehensive thermal management
4. Reduced latency and sufficient security

Pillar 1: Chips and Board Technology

To support AI algorithms, there is an incredible strain placed on chips and boards. AI requires incredibly complex, high computing demands- much more so than normal computers performing more traditional functions. To support those demands, chips and boards must be able to support processing multiple actions simultaneously, multi-language processing and learning functions, process incredible amounts information over extended periods of time with little to no drain on efficiency, speed, energy and bandwidth. The United States is already home to some of the leading chip and board engineers and manufacturers, but needs to invest more in developing expertise and capability. That notwithstanding, even the best and brightest engineers are faced with infrastructure and technology barriers because there is not an environment in the United States to consistently support the chip and board innovation.

Pillar 2: Power Availability & Reliability

For AI dominance, the United States not only needs to have sufficient power in the grid to support the incredible energy consumption needed to run data centers and AI, there needs to be reliability that the energy feeding data centers is available 24 hours a day, 7 days a week. A recent report published by the Department of Energy estimates that data centers in the United States consume nearly 4.5% of the total energy available in the United States. That same report estimates data center energy demands will nearly triple to as high as 12% in the next three years. At the current rate of innovation of chips and boards,

there is a real risk that the energy demands could easily surpass these estimates. Even with the current administration's focus on increasing energy development and production, there is a critical need to operate at optimal energy efficiency. With greater innovation of chips and boards, this need for efficiency will only increase. And, as stated, even if data centers are operating at optimal efficiency, there is still a critical need to ensure that energy is continuously supplied to the data centers. In critical environments, whether its relatively controlled environments like data centers or essential manufacturing facilities, or in the field supporting military or energy mining and production operations, AI cannot perform without a continuous supply of energy.

Pillar 3: Comprehensive Thermal Management

The AI lifecycle, encompassing energy sourcing, data storage, and high-performance computing, relies heavily on robust thermal management. Inadequate cooling at any point can trigger chip and server overheating, resulting in system failures and costly data center downtime. The need for precise temperature control extends to all data centers, not just those supporting AI. Maintaining a consistent temperature range is vital for server, chip, and board performance. To achieve this, data centers employ a range of cooling technologies, from traditional air-cooled systems to cutting-edge direct-to-chip liquid cooling. These solutions utilize various components, including indoor air handling units (CRACs, CRAHs, CDUs, rear door heat exchangers) and outdoor units (dry coolers, adiabatic air handlers, chillers) to ensure uptime, optimize performance, and reduce energy consumption.

Pillar 4: Reduced Latency & Sufficient Security

Securing information is increasingly critical, especially when it involves personal and/or financial information. The United States faces continued threats from domestic and foreign actors who attack our latest technology in an effort to steal information that compromise the safety and stability of US citizens, financial institutions and critical infrastructure, as well as numerous other individuals and institutions. Data is the life blood of AI, as we need clean, accurate data to build, train and validate the large language models. As attacks on our data become more prevalent and sophisticated, the United States needs to be prepared to detect and contain this threat. Developing smaller, more partitioned data center solutions that can work "off-line" is an effective approach that must be explored further. These siloed, off-line data centers limit outside access, while also allowing AI functionality in areas where connectivity to the internet is low or potentially compromised. Moreover, moving high performance compute closer to the point of use reduces time to respond to a given set of inputs, also known as latency. There has been a rise in solutions to move high performance computing to the rugged edge (e.g. conflict regions, mine sites, etc.), while also allowing for high performance computing to be mobile. However, there needs to be more development and adoption in this area. The combination of security needs, offline high performance compute and low latency solutions mandates that the United States continue to develop "edge" data center solutions as an integral part of a more comprehensive AI road map.

Modine Solutions Can Support Growth

Modine's industry leading core competency, providing thermal management solutions, can be a critical component to help address Pillar 3. We have a unique number of data center products that are tied together with a controls platform to offer best in class system efficiency and reliability. For a completely

optimized solution, we believe all pillars must be connected. Specifically, Modine must work closely with the chip and board engineers and manufacturers to understand the characteristics of the dynamic thermal load that needs to be managed. Ensuring proper heat removal from the chips allow them to operate at their full potential at the lowest energy usage. That then ties to power availability and reliability. Every kilowatt of energy that goes into thermal management of high performance compute is a unit of energy that cannot be invested into the solution or adjacent need. For example, imagine high performance compute on a electric combat vehicle operating in stealth mode. The same battery system is needed to propel the vehicle, power the high performance compute needed to achieve success in the overall mission and power the thermal management solution. This last example also highlights the need to reduce latency and have sufficient security. A mobile high performance compute example in a remote conflict region cannot afford to be compromised or wait longer than the enemy to get the optimal answer to a mission critical issue. Modine has traditional data center solutions along with innovative and emerging edge solutions, and has become a trusted partner in making the US a leader in all four pillars. Through this synergistic approach, we are positioned to help ensure America's AI dominance

Conclusion

Modine continues to develop technology through our commercial market channels, but government policy could further encourage development of artificial intelligence solutions. In many areas technology has already been developed, but the entire ecosystem has not been developed yet. Until that ecosystem is better defined and the issues addressed herein are linked, there is a risk that other countries could create policies that encourage this collaboration and build better solutions that become a very credible threat to the United States.